

Predicting Float or Sink: Investigating Density

PhET Density · PhET

Senior Secondary (Years 11-12)

~30 minutes

Science · Chemistry

NOTICE — AI-generated worksheet

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Simulation: <https://phet.colorado.edu/en/simulations/density>

Curriculum links: materials · measurement

Learning intentions

- I can predict and test whether an object floats or sinks based on its density relative to water.
- I can calculate density from measured mass and volume.
- I can evaluate how measurement uncertainty affects a density calculation.

Getting started

Open the simulation and start on the material blocks screen. Before revealing each block's identity, predict whether it will float or sink in water based on its appearance and any information given.

Tasks

1. Choose one material block. Predict whether it will float or sink before releasing it, then release it and record whether your prediction was correct.

2. Note the mass and volume of that block, then calculate its density by hand. Compare your calculated value to any value shown by the simulation.

Working:

3. Repeat the prediction and density calculation for three different material blocks.

Material	Predicted float/sink	Calculated density

4. Using your table, state the rule that determines whether an object floats or sinks in water, in terms of density.

5. Switch to the custom block screen. Build a block with the same mass as one from your table but a different volume. Does it now float or sink? Explain why, even though the mass hasn't changed.

6. Use the custom block screen to find, by trial and adjustment, roughly the minimum volume a 500 g block would need to just float in water. Show your reasoning.

Working:

7. Explain how measurement uncertainty in mass and volume (e.g. from using a balance or a measuring cylinder) could affect the accuracy of a calculated density value, and which measurement you think would introduce more error.

Extension (optional)

Two unlabelled samples have very similar appearance but different densities. Explain how you could use this simulation's approach — measuring mass and volume to calculate density — as a real laboratory method for identifying unknown materials.
